HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and validated low-level hardware drivers in **C** for Qualcomm Snapdragon Subsystems, focusing on **firmware** integration and performance tuning.
- Migrated existing drivers to new platforms, ensuring compatibility and maintaining **validation** integrity across diverse hardware configurations.
- Implemented automation scripts in **Python** to streamline test execution and data collection for hardware validation processes, reducing manual effort by approximately 30%.
- Contributed to the design and execution of **Pre-Silicon and Post-Silicon validation plans**, identifying and documenting critical test cases for SoC subsystems.
- Participated in **code reviews** for driver development, providing constructive feedback to junior engineers to ensure code quality and adherence to standards.
- Managed defect tracking and resolution using **JIRA**, collaborating with hardware designers and verification engineers to debug and resolve complex issues.
- Containerized validation environments using **Docker** and deployed them to **AWS EC2** instances for scalable and reproducible testing.

SKILLS

Languages – C, C++, Assembly, Python, SQL

Processors – ARM, RISC V

Methodologies – Automation, Test Development, Integration, Validation Strategy, Agile

Tools – GIT, Perforce, JIRA, Docker, AWS (EC2, CloudWatch), Jenkins, Splunk, Dynatrace

Domains – Low-level Driver Development, Firmware, System Hardware Validation, Embedded Systems, SoC

PROJECTS

Low-Level Driver Development for Embedded Systems

- Developed and tested **C** drivers for peripheral communication on an embedded Linux platform, integrating with a custom SoC.
- Implemented **firmware** updates and validation routines, ensuring successful deployment and functionality across multiple hardware revisions.
- Authored detailed test plans and executed validation scripts to verify driver stability and performance under various load conditions.

Automation Framework for Hardware Validation

- Created a **Python**-based automation framework to manage and execute test cases for hardware validation, reducing manual test execution time by 40%.
- Integrated the framework with **JIRA** for automated defect logging and reporting, improving issue traceability and team collaboration.
- Designed and implemented data collection mechanisms to capture detailed logs and performance metrics during test runs for post-analysis.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)